

BAZNAS: Empowering Indonesians with transformative Islamic social funds

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ABOUT BAZNAS



BAZNAS absorbed a spike in Ramadan donations, gained insight from millions of data points and facilitated AI-driven business transformation stories with Google Cloud.

About BAZNAS

The National Board of Zakat for the Republic of Indonesia (BAZNAS) is a government agency in Indonesia that is responsible for zakat and other Islamic social funds. Its mission includes enhancing national zakat literacy, maximizing zakat distribution for poverty alleviation and social inequality reduction, and strengthening the national zakat management system with modern, data-driven methods. BAZNAS aims to be a reference for global zakat movements, emphasizing collaboration, professionalism and continuous competence improvement.

Google Cloud results

- Supports mission to spread zakat empowerment with 10 percent improvement in donation collection target
- Deepens charitable impact by absorbing peak cycle demand with zero downtime
- Scales zakat operations by reducing data analytics processing from weeks to hours with BigQuery
- Drives ML-driven business insight with real-time

Abso
surge
cycle
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Industries: Government & Public Sector

Location: Indonesia

(<https://cloud.google.com/>). BigQuery



About PT Mitra Integrasi Informatika

With a vision to be a leading digital economy hub, PT Mitra Integrasi Informatika offers a range of ICT services ranging from system integration to digital transformation and full-scale enterprise resource planning.



Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Compute Engine

(<https://cloud.google.com/compute/>)

Google Cloud Armor

(<https://cloud.google.com/armor/>)

Google Workspace

(<https://workspace.google.com/>)

**crunching of
millions of
nationwide
data points**

transformation and full-scale enterprise
resource planning.

- **Opens new
avenues for
zakat
enablement
with AI-driven
strategy
driven by
Vertex AI**

One of the five pillars of Islam, [zakat](https://www.investopedia.com/terms/z/zakat.asp#:~:text=Zakat%20is%20an%20Islamic%20financial,individual's%20total%20savings%20and%20wealth.)

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is a principle that requires Muslims with means to donate to the less fortunate. For BAZNAS, the Indonesian government body in charge of zakat social funds, the mission goes far beyond charity. It aims to convert recipients into donors through economic, educational and health initiatives.

From helping the poor set up micro-businesses such as roadside fried chicken stalls to providing training in vocations such as plumbing or electrical engineering, BAZNAS pursues a vision of creating virtuous cycles of empowerment.

The mission of collecting and distributing funds across the world's largest Islamic nation, with a population of more than 270 million, is always demanding. Consider how challenging things

become during the holy month of Ramadan, when zakat donations regularly spike five-fold. On its original on-premise architecture, BAZNAS saw its systems crash for hours or even up to a day due to the sudden crush of zakat activity.

Continuously adding services incurred onerous costs. Even then, BAZNAS could not keep up with the burden on its systems, or scale nimbly enough to avoid inefficiencies. "We were at the limit with our bare metal infrastructure, and needed to move to the cloud," says Andrian Johnson, Director, Information Security, Data, and Digital Services at BAZNAS.



BAZNAS initially tried one of the major cloud providers, only to find itself weighed down by the high cost due to the difficulty of monitoring its usage of services. Thanks to a collaborative relationship that was established earlier using Google Workspace (<https://workspace.google.com>) solutions, BAZNAS decided to switch to Google Cloud (<https://cloud.google.com/>). Andrian said that In-depth discussions with the team soon yielded BAZNAS to a whole new cloud experience at affordable pricing.

Partnering with PT Mitra Integrasi Informatika (<https://www.mii.co.id/>), BAZNAS embarked upon a Google Cloud journey built around Compute Engine (<https://cloud.google.com/compute>) for scalable computing, Cloud SQL (<https://cloud.google.com/sql>) for database management and BigQuery (<https://cloud.google.com/bigquery>) for autoscaling data analytics. The migration has resulted in a seamless ecosystem that supports the massive data processing involved in managing zakat across Indonesia.

"We recognized the need for a robust, scalable solution that could handle the demands of zakat collection, especially during peak seasons. Google Cloud emerged as the ideal solution, aligning with our existing use of Google Workspace," says Andrian. "With Google Cloud we're able to absorb any spike in demand seamlessly, even during Ramadan, with infrastructure provisioning in real time to 150 percent of need."

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*—Andrian Johnson,
Director - Information
Security, Data, & Digital
Services, BAZNAS*

Spreading the bounty of zakat empowerment with BigQuery

As BAZNAS lifts Indonesian lives through effective deployment of zakat funds, it finds itself empowering its own team thanks to intuitive solutions such as BigQuery and proactive Google Cloud support.

"In one year, my help desk staffer, who helped people with printers and cables, has become a data

engineer," says Andrian. "Overall, I've been able to upskill my team, adding to my number of data analysts, through Google Cloud training programs. Moreover, BigQuery is just so intuitive that it's easy to begin harnessing the power of advanced data analytics. We went from being data handlers to data strategists."

The simplicity of the BAZNAS Google Cloud ecosystem, in which Cloud SQL as a managed relational database links seamlessly with BigQuery, enables improved data quality for fast operational insight. That bolsters the core mission of efficient zakat collection and distribution.

Currently, BAZNAS runs two BigQuery instances. One serves as a scalable data lake and the other as a data warehouse, driving real-time data intelligence that can be intuitively viewed on Looker dashboards. The advantages of moving to a data analytics protocol driven by BigQuery, from one that relied on manual spreadsheets, has been transformative, says Andrian.

Overnight, BAZNAS gained the ability to pull data from millions of nationwide zakat records in real-time, and query the data warehouse for powerful operational insight. It also gained significant advantages in data quality, through the ability to quickly compare entries from different sources, such as government agencies and charities, check for discrepancies, and rectify problems in one seamless workflow.

With improved data quality and analytics on BigQuery, BAZNAS went from being far below collection benchmarks to being roughly 10 percent above its target for zakat donation collection, achieving more than 900 billion plus rupiah for the BAZNAS HQ Collection alone.

"We are about 10 percent above our target for receiving donations thanks largely to BigQuery data analytics," says Andrian. "Improvement in data quality combines with the ability to crunch data in real time to enable us to make decisions on donation collection quickly and more accurately."

The transformative results have led Andrian to even change the name of his division from "IT unit" to "national digital transformation unit."

"With BigQuery, we're no longer handling just IT, we're basically moving the whole business," says Andrian. "We're moving BAZNAS towards transforming the entire operation through data analytics. Today, data results that previously took weeks to achieve are now coming in within hours due to the power of BigQuery."

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Protecting zakat data integrity with impregnable security

For a program that handles hundreds of millions of sometimes sensitive data points, data security is crucial to success. Not only does Cloud SQL enjoy built-in security features that protect the integrity of

data, says Andrian, it also features the Google Cloud reputational advantage that keeps hackers away before they might even try to strike. In addition, BAZNAS deploys Cloud Armor (<https://cloud.google.com/security/products/armor>) as its intelligent network security system to prevent denial of service and web attacks.

"From Cloud SQL to Cloud Armor, we have several layers of protection that keep our mission of lifting lives and empowering families to the farthest reaches of Indonesia safe from potential data breaches," says Andrian.

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Driving the future of Islamic empowerment with Vertex AI

Looking ahead, BAZNAS is exploring the use of Vertex AI (<https://cloud.google.com/vertex-ai>) to expedite and refine its zakat distribution process. Plans include creating algorithms for fast-tracking donation request assessments and verifying recipient validity remotely.

For example, an orphanage requesting zakat assistance might first be located on Google Maps (<https://mapsplatform.google.com/>). Using Vertex AI to analyze a 360-degree panoramic view of the facility, BAZNAS might be able to assess the applicant for signs of fraud. That would save the agency the effort and cost of visiting the orphanage for physical inspection.



Andrian foresees assessing thousands of zakat proposals per day with Vertex AI, from currently requiring weeks to evaluate a single request. He also anticipates deploying Vertex AI for modeling of whether a zakat program that was effective in one region, such as Western Java, might be viable in another, such as Central Sulawesi.

The next stage of BAZNAS' evolution will also involve working with partner PT Mitra Integrasi Informatika to transition to a services oriented architecture (SOA) designed in a Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine>) microservices framework. This will enable BAZNAS to further streamline its operations through an ecosystem of services oriented apps.

"There's no end to the positive impact we can achieve across Indonesia by elevating our operations with AI insight and a nimble service oriented architecture based on microservices," says Andrian. "Vertex AI and Google Kubernetes Engine promise to become powerful partners in our mission to achieve transformative change through zakat empowerment."

